

We have now covered the basics of kernel-level programming.

Our main goal is to understand how to interact with the kernel and include our own small program within it! Sounds interesting, right? So let's get started!

Creating a *makefile*

The initial step is to create a *makefile*. The 'make' command facilitates us to run a set of commands specified in a *makefile*.

Open a text editor and add the following code. Name it 'Makefile' and save it in a location as shown below.

```
obj-m += jishnuos.o
all:
    make -C /lib/modules/$(shell uname -r)/
```

```
build M=$(PWD) modules
clean:
    make -C /lib/modules/$(shell uname -r)/
build M=$(PWD) clean
```

With the *makefile* ready, we can now write the code to make our kernel run. In this case, we will include a module into our kernel to add two numbers.

Create a file with the name that you have given in the *object* file, as shown in the code here. Now save this file in the same location you have saved your *makefile*.

```
#include<linux/module.h>
#include<linux/kernel.h>
#include<linux/kthread.h>
#include<linux/sched.h>
#include<linux/time.h>
```

```
int init_module(void)
{
    int pi
    int q = 5;
    int r = 3;
    p= q + r;

    printk (KERN_INFO "The sun of the
    numbers is %d\n",p)
    return 0;
}

void cleanup_module(void){
    printk(KERN_INFO"BYE\n");
}
```

Once the kernel program is ready, we can insert this module into the kernel.

To do this, first run the 'make' command. Open a terminal and enter the 'make' command as shown in Figure 2.

Next, open a terminal, go to the location where you have stored the files and use the command as shown in Figure 3.

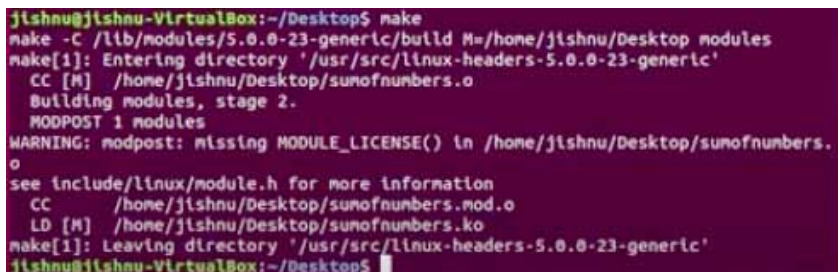
```
sudo insmod sumofnumber.ko
```

Now that our module is inserted, we can use the 'dmesg' command to check our output, as shown in Figure 4. It displays the sum of the two numbers we provided, indicating that our code has worked!

We have successfully inserted our own module into the kernel, which is pretty amazing. You can explore further with this as the base; the deeper you go, the more fun it gets! You can create and develop your own projects and implement new functionalities into Linux. Who knows, your work may even be accepted by Ubuntu itself! 

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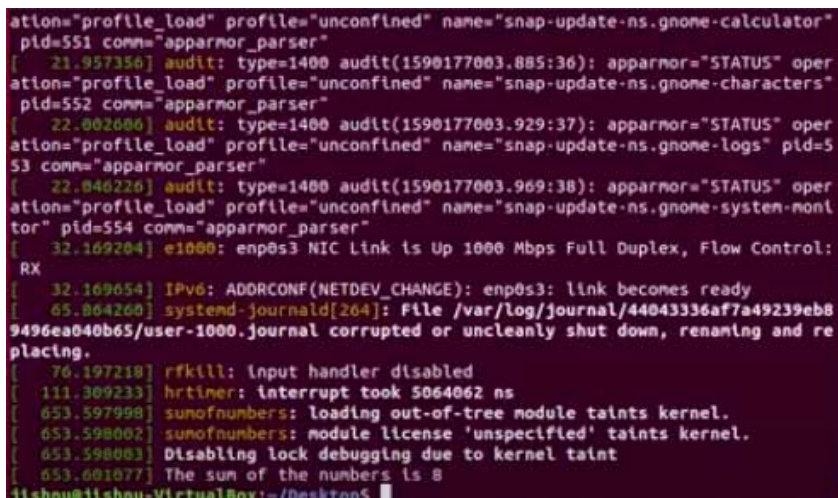
```
jishnu@jishnu-VirtualBox:~/Desktop$ make
make -C /lib/modules/5.0.0-23-generic/build M=/home/jishnu/Desktop modules
make[1]: Entering directory '/usr/src/linux-headers-5.0.0-23-generic'
CC [M] /home/jishnu/Desktop/sunofnumbers.o
Building modules, stage 2.
MODPOST 1 modules
WARNING: modpost: missing MODULE_LICENSE() in /home/jishnu/Desktop/sunofnumbers.o
see include/linux/module.h for more information
CC /home/jishnu/Desktop/sunofnumbers.mod.o
LD [M] /home/jishnu/Desktop/sunofnumbers.ko
make[1]: Leaving directory '/usr/src/linux-headers-5.0.0-23-generic'
jishnu@jishnu-VirtualBox:~/Desktop$
```

Figure 2: *make* command



```
[sudo] password for jishnu:
jishnu@jishnu-VirtualBox:~/Desktop$ sudo insmod jishnuos.ko
jishnu@jishnu-VirtualBox:~/Desktop$
```

Figure 3: *insmod* command



```
ation="profile_load" profile="unconfined" name="snap-update-ns.gnome-calculator"
pid=551 comm="apparmor_parser"
[ 21.957356] audit: type=1400 audit(1590177003.885:36): apparmor="STATUS" oper
ation="profile_load" profile="unconfined" name="snap-update-ns.gnome-characters"
pid=552 comm="apparmor_parser"
[ 22.002006] audit: type=1400 audit(1590177003.929:37): apparmor="STATUS" oper
ation="profile_load" profile="unconfined" name="snap-update-ns.gnome-logs" pid=5
53 comm="apparmor_parser"
[ 22.040226] audit: type=1400 audit(1590177003.969:38): apparmor="STATUS" oper
ation="profile_load" profile="unconfined" name="snap-update-ns.gnome-system-monit
or" pid=554 comm="apparmor_parser"
[ 32.169204] e1000: enp0s3 NIC Link is Up 1000 Mbps Full Duplex, Flow Control:
RX
[ 32.169654] IPv6: ADDRCONF(NETDEV_CHANGE): enp0s3: link becomes ready
[ 45.804260] systemd-journald[264]: File /var/log/journal/44043336af7a49239eb8
9496ea040b65/user-1000.journal corrupted or uncleanly shut down, renaming and re
placing.
[ 70.197218] rtkill: input handler disabled
[ 111.309233] hrtimer: interrupt took 5064062 ns
[ 653.597998] sunofnumbers: loading out-of-tree module taints kernel.
[ 653.598002] sunofnumbers: module license 'unspecified' taints kernel.
[ 653.598003] Disabling lock debugging due to kernel taint
[ 653.601077] The sum of the numbers is 8
jishnu@jishnu-VirtualBox:~/Desktop$
```

Figure 4: Final output